



Onsite environmental analysis by portable mass spectrometry

Jianfeng Zhang^a, Zhen Zhou^{a,*}, Bin Hu^{a,b,**}

^a College of Environment and Climate, Institute of Mass Spectrometry and Atmospheric Environment, Guangdong Provincial Engineering Research Center for On-line Source Apportionment System of Air Pollution, Jinan University, Guangzhou, 510632, China

^b Guangdong Provincial Key Laboratory of Speed Capability Research, Jinan University, Guangzhou, 510632, China

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ABSTRACT

Onsite analysis is essential for the rapid determination of the chemical composition of environmental samples in timely decision-making in environmental monitoring and assessment, and thus has attracted considerable interest in developing portable analytical instruments. Recently, versatile portable mass spectrometry (MS) approaches have been developed as promising analytical tools for onsite analysis of environmental samples, due to their portability, miniaturized design, and excellent analytical performances. This review aims to introduce the techniques and principles of portable MS approaches, emphasizing their features, advantages, and limitations. New sampling methods, sample preparation approaches, and ionization techniques of portable MS for onsite analysis are highlighted. The significant applications of portable MS in onsite analysis of pollutants in air, water, and soil, as well as in emergency environmental monitoring and rapid response to pollution incidents, are reviewed. The prospects for further development and potential applications are also discussed.

1. Introduction

Various pollutants can be released into the air, water, and soil from various complex sources such as industrial emissions, vehicle exhaust, and agriculture activities, which bring complex harms to ecology, environment, and human health [1–3]. In dangerous chemical accidents, e.g., leaks, fires, or explosions, hazardous pollutants could be suddenly generated and rapidly diffused into environments [4–6]. Therefore, reliable, rapid, accurate, and comprehensive information on chemical compositions and their concentrations, distributions, and impacts at contaminated sites is crucial for assisting in pollution control, emergency management, and environmental assessment [7,8]. The development and application of onsite analytical tools is highly needed in timely identification and quantification of pollutants in field environments. Portable samplers and analytical tools offer significant advantages for onsite environmental analysis by eliminating time delays, minimizing sample consumption, and enabling a better understanding of pollutant compositions and their characteristics such as distribution, diffusion, and transformation.

Onsite analysis of environmental samples typically requires state-of-the-art techniques for sample collection and preparation, analyte identification and quantitation. Current techniques for the chemical analysis

of environmental samples encompass a variety of analytical platforms, including sensors, spectroscopy, chromatography, and mass spectrometry [9–11], which facilitate the rapid detection, characterization and quantification of analytes in complex field environments. Among these analytical techniques, mass spectrometry (MS) has been widely employed as a powerful tool for the precise analysis of environmental samples due to its high sensitivity, excellent specificity, and rapid signal response [12–16]. It is a fact that most MS methods mainly focus on laboratory analysis of environmental samples, providing comprehensive information on the pollutants. However, sample preparation processes such as collection, storage, transport, extraction, and separation are typically required prior to MS analysis. These procedures are often time-consuming and labor-intensive in the laboratory. Therefore, onsite environmental analysis often lacks sufficient timeliness, especially during pollution incidents. Moreover, many chemically active pollutants may undergo transformation, degradation, or reactions during sample preparation, leading to a loss of representativeness in the analytical results [7]. To address these challenges, the development of portable MS approaches integrating direct sampling/ionization techniques represents a significant advance in analytical chemistry. Portable MS traditionally refers to instruments that are compact and lightweight enough to be carried by a single person [17] and operated directly in field

* Corresponding author.

** Corresponding author. College of Environment and Climate Jinan University Guangzhou, 510632 China.

E-mail addresses: zhouzhen@gig.ac.cn (Z. Zhou), bin.hu@jnu.edu.cn (B. Hu).

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environments. Therefore, portable MS instruments offer a rapid, simplified, and convenient approach for onsite environmental analysis, enabling real-time insights into environmental processes and opening up new possibilities for field-based monitoring and decision-making. Various portable MS platforms have been developed for diverse environmental applications and continue to undergo further refinement and optimization. The integration of novel sampling methods with portable MS enables the effective analysis of complex matrices in environments [18–21].

In recent years, extensive research efforts have been devoted to the development and application of portable MS techniques for onsite analysis in various fields. Several excellent review articles have comprehensively covered the technical evolution of portable MS systems and their broad applications in areas such as food safety analysis and clinical diagnostics [12,22–27], there remains a gap in critically reviewing onsite environmental analysis, which is one of significant applications driving the development of portable MS. In particular, the integration of emerging direct sampling and ionization strategies with portable MS for rapid detection of environmental pollutants in field environments has not been sufficiently addressed. To fill this gap, the present review focuses on onsite analysis of environmental samples by portable MS, emphasizing how recent advancements have simplified analytical workflow to as few as one or two steps, due to great improvements of sampling/ionization methods. The current capabilities of portable MS for analyzing diverse environmental matrices, such as air, water, and solid samples, and for application in emergency environmental monitoring and rapid response to pollution incidents are summarized. Finally, prospects for further development and future applications of portable MS in environmental monitoring are also explored.

2. Portable mass spectrometry approaches

To date, numerous MS approaches have been miniaturized, with several achieving true portability for onsite environmental analysis. Representative examples include gas chromatography-mass spectrometry (GC-MS), membrane inlet mass spectrometry (MI-MS), ambient ionization mass spectrometry (AI-MS), and micro-electromechanical system mass spectrometer (MEMS-MS). These approaches are categorized according to their key features, namely sampling method (e.g., GC-MS, MI-MS), ionization technique (e.g., AI-MS), or construction design (e.g., MEMS-MS). Beyond these categories, several specialized portable

MS designs have also been reported, such as portable pyrolysis MS [28], portable double-focusing MS [29], portable nano-liquid chromatography-electron ionization MS (nanoLC-EI-MS) [30], and palm portable MS [31]. The key parameters of these systems, including size, weight, power consumption and battery duration, as well as technical characteristics (sampling method, ionization technique, and mass range), are summarized in Table 1. It should be noted that continuous data acquisition can be employed in real-time monitoring using MI-MS and AI-MS with continuous atmospheric pressure interfaces, which enable time-resolved real-time monitoring. In contrast, portable MS platforms that rely on sample pretreatment processes such as portable GC-MS and AI-MS with discontinuous atmospheric pressure interfaces are typically operated in a discontinuous acquisition mode that generates non-continuous monitoring.

2.1. Portable gas chromatography-mass spectrometry

Portable gas chromatography-mass spectrometry (GC-MS) is primarily used for onsite detection of volatile organic compounds (VOCs) [32]. One major advantage of portable GC-MS is the use of a short GC column, which enables rapid separation of complex VOC mixtures for MS detection, typically performed using a standard electron ionization (EI) source (Fig. 1a) [33–36]. Various VOC sampling methods have been developed for coupling with portable GC-MS in diverse environmental applications (Fig. 1a). Among them, air injectors are commonly used to directly collect gaseous samples from the atmosphere or air bags, whereas liquid injectors are applied for collecting liquid samples. These injectors provide a simple means of introducing VOCs into the GC-MS for onsite analysis. However, injectors often suffer from sample loss, small sample volumes, and poor sensitivity when detecting trace VOCs in complex environments. To address these limitations, enrichment techniques such as purge and trap (P&T) [37], headspace solid-phase microextraction (HS-SPME) [18], liquid-phase microextraction (LPME) [38], and low-temperature adsorption thermal desorption (LT-ATD) [39] are typically required before portable GC-MS analysis. Particularly, SPME is a powerful sample pretreatment technique capable of selectively enriching VOCs from complex environmental matrices using sorbent materials, and can be directly inserted into portable GC-MS for analyte desorption and detection without any hardware modification. This makes SPME a simple, convenient and highly efficient sampling method for portable GC-MS. Moreover, SPME can be used not only for gaseous samples but also for water and soil matrices. In recent years,

Table 1
Typical portable MS approaches.

Mode	Type	Ionization	Size, cm	Weight	Range	Power	Duration	Ref
Field GC-MS	GC-MS	EI	/	90 kg	12–400 Da	250 W	/	[33]
Guardion-7	GC-MS	EI	47 × 36 × 18	<13 kg	50–442 m/z	80 W	/	[35]
Tridion-9	GC-MS	EI	38 × 39 × 23	14.5 kg	45–500 amu	/	2.5 h	[36]
GC-MS 2000	GC-MS	EI	43 × 34 × 20	19 kg	15–550 amu	/	2 h	[42]
EXPEC 3500	GC-MS	EI	44 × 42 × 26	<19 kg	40–450 amu	/	/	[40]
HAPSITE® ER	GC-MS	EI	46 × 43 × 18	19 kg	45–300 m/z	/	/	[118]
Griffin 450™	GC-MS	EI	48.8 × 48.8 × 53.6	44 kg	/	600 W	/	[34]
Portable GC-LIT-MS system	GC-MS	EI	43 × 35 × 20	18 kg	15–550 amu	/	/	[37]
miniRuedi	MI-MS	EI	60 × 40 × 14	13 kg	/	50 W	/	[9]
Portable MS system	MI-MS	EI	61.6 × 49.3 × 22	17 kg	1–500 m/z	24 W	/	[48]
Mini 10	MI-MS	EI, GDEI	32 × 22 × 19	10 kg	Over 500 m/z	70 W	1 h	[46]
Mini 11	MI-MS	GDEI, ESI, APCI, DESI	22 × 12 × 18	5 kg	Up to m/z 1500	35 W	2 h	[47]
SPI-TOFMS	MI-MS	SPI	30 × 20 × 20	15 kg	/	<100 W	3 h	[49]
DT-100	MI-MS	SPI	49 × 39 × 16	18 kg	30–300 Th	60 W	4 h	[50]
Mini 12	AI-MS	APCI, PSI, nESI	49.8 × 56.1 × 41.9	25 kg	Up to m/z 900	<100 W	/	[75]
Mini 14	AI-MS	PSI, nESI	26.5 × 26.5 × 37	12 kg	Over 2000 m/z	<100 W	3 h	[76]
Mini-MS (Cell)	AI-MS	PSI, EESI, nESI	33 × 23 × 15	<10 kg	50–1000 Da	80 W	/	[69]
Mini-MS	AI-MS	APCI, PSI, nESI	30 × 30 × 18	6 kg	200–2500 Da	/	/	[65]
MT Explorer 100	AI-MS	nESI, AP-MALDI, APCI	48 × 43 × 42	68 kg	30–2500 m/z	500 W	/	[78]
Mini S	AI-MS	LTP	30.5 × 29.5 × 12.5	10 kg	100–925 amu	65 W	1.5 h	[77]
Palm portable MS	Direct MS	EI	8.2 × 7.7 × 24.5	1.48 kg	45–300 m/z	5 W	10 h	[31]
Py-MS	Direct MS	EI	70 × 30 × 40	18 kg	1–300 amu	180 W	4 h	[28]

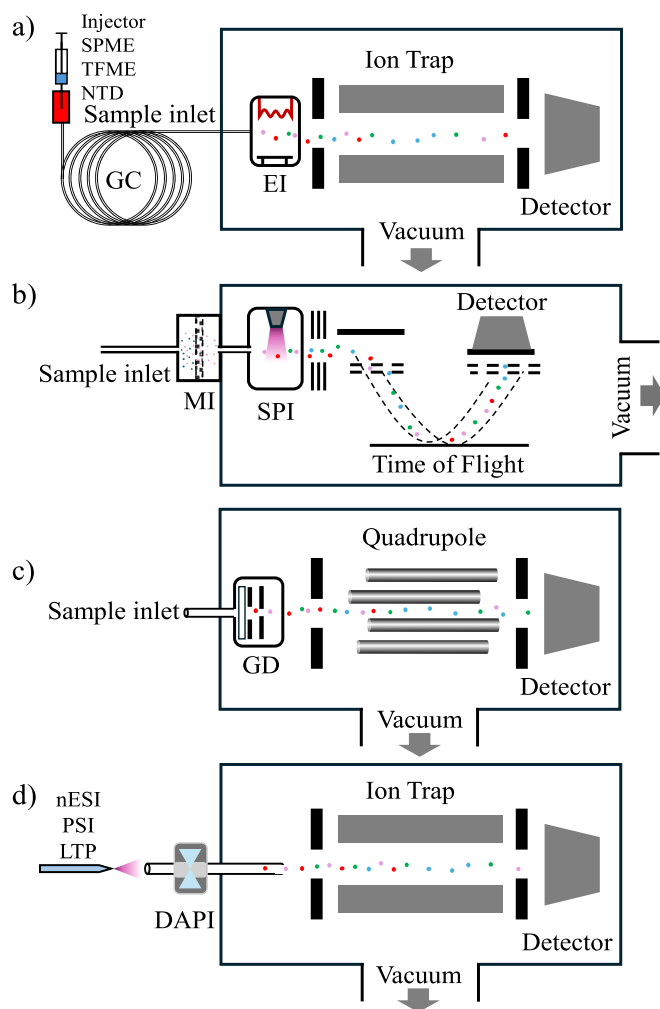


Fig. 1. Schematic illustration of the main components of portable MS approaches: a) GC-MS, adapted from Ref. [36] with permission; b) MI-MS, adapted from Ref. [49] with permission; c) MEMS-MS, adapted from Ref. [58] with permission; d) AI-MS, adapted from Ref. [69] with permission.

different forms and variants of SPME techniques, *e.g.*, thin film micro-extraction (TFME) and needle trap device (NTD), have been developed for coupling with portable GC-MS [40,41]. And the integration of SPME-based techniques with the robotic platforms, *e.g.*, drones, unmanned aerial vehicle, and bio-robot, further expands their potential for sampling VOCs from hazardous or inaccessible environments for onsite portable GC-MS analysis [42–44]. Portable GC-MS also provides the advantage of spectral matching with the NIST library, enabling identification of unknown compounds and reducing false positives. However, due to the reliance on offline sampling and GC separation, portable GC-MS is not suitable for real-time VOC monitoring.

2.2. Portable membrane inlet mass spectrometry

Portable membrane inlet mass spectrometry (MI-MS) utilizes a thin membrane as an interface between the sample flow and the MS inlet (Fig. 1b), enabling analytes in the gas or liquid phase to selectively permeate through the membrane and transfer to the MS for analysis [45–50]. In-vacuum ionization sources such as electron ionization (EI), photoionization (PI), and chemical ionization (CI) are typically used in portable MI-MS [49,51]. PI and CI are soft ionization techniques, offering advantages for the generation and identification of analytes in field environments. Portable MI-MS maintains vacuum environment by a pumping system, where a membrane is used for VOC introduction but

prevents the entrance of water and most inorganic gases from sample flow to MS system. The membranes used are mainly hydrophobic and semi-permeable polymer materials, enabling selective and sensitive detection of VOCs and SVOCs from raw samples. To enhance VOC detection, various membrane materials such as capillary hollow fiber membranes (HFMs) and polydimethylsiloxane sheet membranes (PDMS) [52–54] have been explored. The geometric and physico-chemical properties of membrane materials are critical factors influencing VOC detection in MI-MS. By integrating online sampling/ionization, MI-MS approach eliminates sample pretreatment and separation steps, further enhancing its efficiency for environmental applications [55,56].

2.3. Portable micro-electromechanical system mass spectrometer

Micro-electromechanical systems (MEMS) have driven the development of highly miniaturized mass spectrometry in the past three decades [57]. A distinctive feature of MEMS-MS is its high level of integration into a chip (MS-on-a-chip), which allows it plays a role between traditional chemical sensors and current miniature MS systems (Fig. 1c) [58]. MEMS-MS is expected to become a powerful portable instrument for onsite environmental analysis. It is a microsystem that integrates mechanical components, ion source, analyzer, sensors, actuators, and electronics using microfabrication technologies [59–61]. Various types of mass analyzers, such as ion traps, quadrupoles, Wien filter, and time-of-flight (TOF) analyzers, have been successfully micromachined [62]. The EI source is the primary ionization method used in MEMS-MS, as it can be fabricated at a very small size. Microplasma ionization, derived from common plasma technologies such as glow discharge, dielectric barrier discharge, microwave-induced plasma, capacitively coupled plasma, and inductively coupled plasma, is considered a soft ionization method for MEMS-MS systems. Conventional ESI and photoionization are expected to be microfabricated and incorporated into MEMS-MS systems. Despite the significant breakthroughs and new possibilities of MEMS-MS, so far, a fully functional MEMS-MS is not yet commercially available. Further technology development of MEMS-MS is needed for rapid growth in onsite environmental analysis.

2.4. Portable ambient ionization mass spectrometry

Direct MS analysis with ambient ionization (AI) and atmospheric pressure interface (API) ionization has emerged as a revolutionary analytical technique, enabling the direct desorption/ionization of analytes on benchtop MS systems with minimal sample preparation under atmospheric conditions [63,64]. AI and API technologies are operated under atmospheric pressure, which poses a great challenge to the vacuum system of MS analyzer. The continuous atmospheric pressure interface (CAPI) has been widely used in conventional laboratory benchtop MS systems due to its robustness and compatibility with various API and AI sources. However, because CAPI require large pumps and complex electronic systems with high power consumption, it is difficult to integrate into portable MS systems. In 2015, Xu and co-workers introduced the first generation mini-MS with a CAPI interface, utilizing a differential pumping system with a two-stage vacuum manifold [65]. This advancement enabled the development of mini-MS systems incorporating API (*e.g.*, nESI) and AI (*e.g.*, paper spray) techniques [65–68]. A key limitation in interface design is the trade-off between ion transfer and vacuum demands: larger orifices improve ion transfer but increase neutral gas influx, requiring stronger pumps; smaller orifices reduce gas load but limit ion transmission to the analyzer. Another remarkable advancement in portable AI-MS is the development of the discontinuous atmospheric pressure interface (DAPI) [12], where the ion transfer tube is rapidly and intermittently opened and closed, enabling ions generated under ambient conditions to be efficiently introduced into the MS for rapid analysis of raw samples (Fig. 1d) [69]. Although DAPI results in discontinuous MS signal

acquisition, mini-MS systems can still achieve approximate real-time monitoring at second level. A wide range of portable MS methodologies have been developed using API techniques (e.g., APCI, nESI) and AI techniques (e.g., paper spray ionization, wooden-tip ESI, LTP, EESI, and AP-MALDI) for direct analysis of complex samples [22,65,69–79]. Compared to vacuum-based ionization techniques such as EI and PI, which are mainly limited to small volatile molecules (< 500 Da), the development of portable AI-MS has expanded the accessible mass-to-charge (m/z) range up to 2500 Da or more, which significantly broadens the scope of detectable compounds, including non-volatile organic analytes, metal complexes, organometallic compounds, and biological molecules (e.g., peptides, proteins). Direct coupling portable MS with versatile AI techniques not only expand the possibilities for onsite analysis of environmental analytes in gaseous, liquid, and solid forms without the need for sample transport to a laboratory, but also promote the development of eco-friendly methods and green analytical chemistry in environmental applications [80]. Currently, portable miniature AI-MS technologies are continuously and rapidly evolving [81], holding great promise for improving analytical efficiency and effectiveness in real-world environmental applications.

2.5. Others

Considerable efforts have long been devoted to developing diverse portable MS configurations to overcome the limitations of conventional designs in environmental applications. A portable MS device based on a miniature double-focusing sector-field mass spectrometer has been developed for in situ gas analysis and atmospheric species monitoring in field applications [29], with its development traceable to early studies [82]. Nevertheless, owing to technical limitations and performance constraints, the environmental applications of portable double-focusing MS remain scarce. A novel portable pyrolysis-mass spectrometry (Py-MS) system has been developed to eliminate the complex extraction and separation steps that are typically required in pyrolysis/thermogravimetric-GC-MS (Pyr/TGA-GC-MS) [28]. Py-MS enables rapid monitoring of microplastics based on the characteristic mass spectra of pyrolyzed polymer products. Abonamah et al. [30] demonstrated the first field-portable nanoLC-EI-MS as a proof of concept. The system was transported to the field and tested using a low-power (1500 W) portable generator. Mixture samples were separated, analyzed, and identified in less than 10 min, showing great potential for onsite environmental applications. Yang et al. [31] developed a palm-sized portable MS characterized by low weight, compact size, and low power consumption. Equipped with a miniaturized ion trap and an ion-getter pump, and operating in pulse mode for gas introduction, this device offers high portability and is suitable for integration with unmanned robotic technologies. However, its analytical performance and practical applicability still require further improvement.

It is worth noting that the vacuum system represents one of the most cost- and energy-intensive components in portable MS, with its design profoundly impacting achievable vacuum levels, sample introduction efficiency, instrument size, and overall analytical performance. As such, it remains a pivotal and technically challenging element in the miniaturization of MS devices [81]. Unlike conventional laboratory-scale instruments that rely on large, heavy, multistage vacuum pumps (often exceeding 40 kg), portable MS platforms employ compact, energy-efficient vacuum systems. Typically, these systems integrate a roughing pump stage, commonly implemented using diaphragm or scroll pumps, coupled with turbomolecular pumps for high-vacuum operation to achieve working pressures in the high-vacuum range. Early miniaturization approaches explored the removal of rough pumps by utilizing lightweight ion pumps; however, this strategy constrained sample introduction to pulsed gas injections, thereby limiting continuous operation capabilities [31]. Currently, most portable MS instruments utilize two-stage vacuum configurations that optimize the trade-off between system size, power consumption, and analytical

performance. However, considering factors such as size, weight, energy consumption, cost, and durability, the adoption of a single integrated pump is highly attractive. Therefore, the future development of micro vacuum systems is expected to focus on implementing this single integrated pump solution.

3. Onsite analysis of environmental samples

Portable MS approaches have been widely employed as a powerful analytical tools for onsite environmental analysis, enabling rapid and reliable detection of pollutants in air, water, soil, and other environmental matrices (Table 2). These diverse applications, ranging from air quality monitoring and water testing to soil analysis and emergency response, facilitate timely data collection and support informed decision-making for pollution control and environmental management.

3.1. Air monitoring

It is reported that approximately 99% of the global population currently breathes air that exceeds WHO air quality guidelines. Air pollution has become a global problem [83,84]. Air pollutants including volatile organic compounds (VOCs), fine particulate matter, sulfur oxides (SO_x), nitrogen oxides (NO_x), and others continue to rise from complex emission sources. These pollutants also undergo transformation and dispersion under varying spatiotemporal, geospatial, and environmental conditions [85]. Rapid and accurate measurement of air pollutants is crucial for understanding their pollution characteristics and for managing their emissions effectively [86–88]. Various portable MS approaches have been used for onsite monitoring of air pollutants. For example, MI-MS has been employed to qualitatively and quantitatively detect volatile chlorinated hydrocarbons, polycyclic aromatic hydrocarbons, halogenated hydrocarbons, organosulfur compounds, and other air pollutants, serving both air quality assessment and industrial emissions monitoring purposes [56,89–92]. These studies showing the feasibility of onsite detection of low molecular weight VOCs. Mach et al. [93] utilized portable MI-MS to onsite analyze BTEX compounds (benzene, toluene, ethylbenzene, and xylene) in the remote region in sub-Antarctic Puerto Williams, Chile. This field expedition underscores the great advantages of portable MI-MS for onsite environmental analysis in a remote region, providing scientific evidence of anthropogenic contamination. In another example, portable mini-MS equipped with atmospheric pressure chemical ionization (APCI) have been developed for real-time monitoring of toxic compounds in ambient air [94–96]. These systems enable rapid detection, identification, and quantification of both organic and inorganic gaseous compounds, achieving short analysis times of less than 5 s. To perform air monitoring at high altitudes, drone-based MI-MS platform has been reported for online monitoring and mapping of hazardous VOCs across different airspaces along with spatiotemporal, geospatial, and environmental information [97]. This approach offers a unique advantage by enabling simultaneous online air sampling/ionization, facilitating real-time acquisition of diffusion and distribution patterns of targeted VOCs (Fig. 2a). Despite these advancements in direct air sampling for online monitoring, portable MS approaches still face challenges in detecting low concentrations of VOCs in ambient air.

A rapid pretreatment process (e.g., separation, enrichment) is essential for air analysis, especially for ultra-trace analytes. Onsite SPME coupled with portable GC-MS offers efficient sample collection and high apparatus portability. Barreira et al. [19] investigated the applicability of dynamic SPME coupled with portable GC-MS for the analysis of atmospheric bio-VOCs, including monoterpenes, their oxidation products, and amines, which were efficiently identified. By comparing total monoterpene concentrations obtained via SPME-PGC-MS with those from PTR-MS, a correlation was observed, demonstrating the effectiveness and reliability of SPME-PGC-MS for onsite VOCs analysis. A handheld heated probe coupled with portable GC-MS has also been

Table 2

Portable MS for onsite analysis of various environmental samples.

Samples	Portable MS	Analytes	Sampler	LOD	RSD	Refs
Air	GC-MS	8 chlorinated-VOCs	Hand-held heated probe	0.3–0.7 $\mu\text{g}/\text{m}^3$	/	[98]
Air	GC-MS	3 hazardous VOCs	NTD	0.37–6.99 ng/mL	<20%	[43]
Air	GC-MS	7 PAHs, 8 VOCs	NTD	0.17–0.65 ng/mL (PAHs); 0.2–0.75 ng/mL (VOCs)	<11%	[41]
Air	GC-MS	19 VOCs	SPME	39–136 ng/L	3.11–19.28%	[42]
Air	MI-MS	9 toxic industrial compounds	Thermal desorption inlet	0.0008–3 ppmv	/	[92]
Air	MI-MS	4 chlorinated hydrocarbons	Membrane sampling probe	2.41–7.34 ppb	/	[90]
Air	MI-MS	12 halogenated hydrocarbons	Membrane inlet sampling probe	1.41–8.92 ppb	<5%	[89]
Air	MI-MS	8 volatile organosulphur compounds	Membrane inlet sample probe	0.506–1.257 ppb	<4%	[91]
Air	AI-MS	3 benzene series	Small sample pump	0.2–0.7 ppb	7.6–12.4%	[95]
Air and water	MI-MS	10 pollutants	Gas pump (gas); Water pump (water)	0.5–20 ppbv (gas); 0.5–50 ppb (water)	/	[56]
Water	GC-MS	10 PCBs-OCs; 8 PAHs	SPME	1.43–21.50 ng/L (PCBs-OCs); 0.32–0.62 ng/L (PAHs)	6.3–15.5%	[106]
Water	GC-MS	10 pesticides	Drill-thin film microextraction	1.0–30 ng/L	1–21%	[38]
Water	GC-MS	6 munitions constituents	SPE	0.2–1.4 $\mu\text{g}/\text{L}$	/	[34]
Water	MI-MS	10 VOCs	Water pump	0.31–1.26 ng/mL	2.93–11.03%	[108]
Water	MI-MS	3 benzene series VOCs	Membrane sample probe	4.88–7.46 $\mu\text{g}/\text{L}$	1.56–2.60%	[104]
water	MI-MS	2 benzene series VOCs	Water pump	200–400 ppb	/	[103]
Water	MI-MS	10 VOCs	Membrane sampling probe	69–375 ppb	/	[105]
Soil	GC-MS	5 benzene series VOCs	SPME	100–200 $\mu\text{g}/\text{m}^3$	10.5–15.3%	[20]
Soil	GC-MS	Trichloroethylene	SPME	100 $\mu\text{g}/\text{m}^3$	/	[114]
Soil	GC-MS	59 organic contaminants	Coiled microextraction	0.5–4 ppm	/	[116]
Soil	GC-MS	Polychlorinated biphenyls	SPME	10 ppm	/	[113]
Soil	MI-MS	6 monoterpenes and monoterpene alcohols	A sorbent sampling probe	10 $\mu\text{g}/\text{m}^3$	/	[112]

employed to improve the detection of hazardous VOCs in indoor environments [98]. In particular, drone-based SPME arrays [42] and robot-based NTD arrays [43] have also been developed for remote sampling of hazardous VOCs in different airspaces or dangerous places for onsite analysis by portable GC-MS. Although such remote sampling and preconcentration sacrificed a little bit of analysis time (usually less than 5 min), they significantly improve detection sensitivity and broaden the applicability of portable GC-MS.

Onsite analysis of air pollutants by portable MS offers effective solutions for air pollution control. However, most current studies focus on VOCs analysis in outdoor or low-altitude environments (usually less than 1.0 km). Onsite MS analysis of particulate matter, SO_2 , and NO_x still remains challenging in many aspects, including sampling, ionization, and identification. Future developments in portable MS are expected to further enhance its performance, improving detection sensitivity, expanding the range of detectable air pollutants, and increasing overall reliability in field measurements.

3.2. Water quality analysis

Water is a crucial source for human health and ecological system. However, contamination from natural and anthropogenic sources, such as hazardous chemicals, heavy metals, pesticides, microplastics, and emerging contaminants, poses a significant threat to public health worldwide [99,100]. Timely and reliable water quality assessment requires near real-time onsite analysis to inform rapid decision-making. Portable MS approaches play a critical role in the onsite detection and characterization of different pollutants in environmental water. For example, a Py-MS system has been developed for the rapid identification and quantification of microplastics, allowing microplastics decomposition within a compact pyrolyzer and generating chemical fingerprints within 5 min [28]. Mächler et al. [101] designed a stand-alone system based on portable gas equilibrium (GE)-MIMS for onsite dissolved gas analysis (He, Ar, Kr, N_2 , and O_2) in groundwater, achieving simultaneous and quasi-continuous measurements. Other portable MI-MS system equipped with a heat exchanger interface [102], a water sampling pump [103], or a membrane sample probe [104,105] have been employed for online monitoring of VOCs in water, reaching detection

limits at $\mu\text{g}/\text{L}$ level.

To enhance sensitivity for trace pollutants in environmental water, sample pretreatment is often necessary before MS analysis. Hu et al. [106] integrated SPME coupled with portable GC-MS for onsite detection of polychlorinated biphenyls, organochlorine pesticides, and polycyclic aromatic hydrocarbons, enabling extraction, detection, and identification of all analytes within 30 min. Liu et al. [69] further advanced a biocompatible SPME and a nESI emitter into a kit for coupling with a miniature MS to enable onsite microextraction and analysis of pollutants from high-salt seawater, wastewater, and aquatic organisms. A key feature of the bio-SPME-nESI-mini-MS system is its capability for onsite analysis of non-volatile pollutants, thereby broadening in water analysis. More recently, compact capillary LC systems have been successfully coupled with portable MS, enabling onsite quantification of haloacetic acids in water with detection limits below regulatory levels and minimal sample preparation, highlighting the potential of liquid-phase separations in field-based monitoring [107].

To improve the safety in handling hazardous water samples, unmanned robotic technologies have been successfully combined with portable MS for remote water pollutant sampling and detection. Drone-based TF-SPME was developed for remotely extracting and storing hazardous VOCs for onsite detection by portable GC-MS (Fig. 2b) [44]. In this approach, the drone was remotely controlled to reach the designated site for VOC sampling, followed by contamination analysis using portable GC-MS. Zhang et al. [108] developed a novel unmanned water analysis system integrating a portable MI-MS with an automatic sampling device on a remote-controlled unmanned ship for real-time online both qualitative and quantitative detecting of hazardous VOCs in environmental water, enabling navigation and compositional analysis of VOCs in situ (Fig. 2c). Furthermore, this research rigorously optimized key experimental parameters to achieve optimal system performance, showing low limits of detection for 10 VOCs (LODs: 0.31–1.26 ng/mL), good reproducibility (RSDs: 2.93%–11.03%), and excellent quantitative capability ($R^2 > 0.99$). These analytical performances show the potential of robotic integration to enable efficient, reliable, and safe in situ monitoring of toxic water, supporting broader deployment in environmental fields.

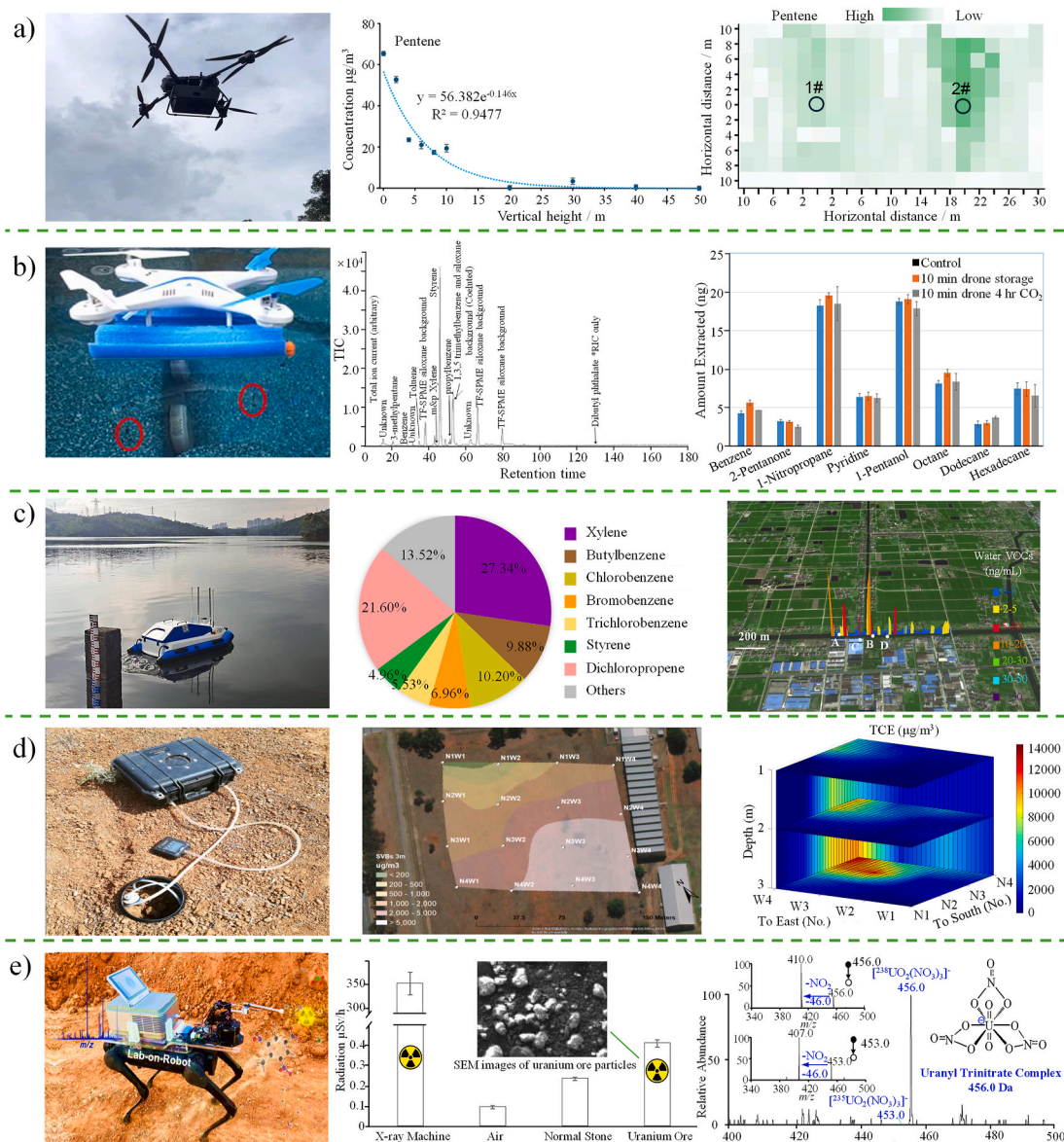


Fig. 2. Portable MS analysis of complex environmental samples: a) drone-based MI-MS for online monitoring of air samples, reproduced from Ref. [97] with permission; b) drone-based TF-SPME coupled with portable GC-MS for water detection, reproduced from Ref. [44] with permission; c) unmanned ship-based MI-MS for online water monitoring, reproduced from Ref. [108] with permission; d) soil vapor sampling probe coupled with portable GC-MS for soil TCE detection, reproduced from Ref. [114] with permission; and e) mini MS-on-robot system for onsite analysis of soil mineral elements and their radioactive isotopes, reproduced from Ref. [71] with permission.

3.3. Soil contaminant analysis

Soil matrices are very complex, containing various components such as moisture, salts, organic matter, and inorganic matter, which can interfere with accurate detection and impose high demands on onsite MS analysis [109–111]. Advanced sample preparation and ionization methods are expected to help overcome these challenges. For instance, a sorbent sampling probe coupled with MI-MS has been used for the detection of volatile monoterpenes and monoterpene alcohols in forest soil atmosphere [112]. HS-SPME coupled with portable GC-MS play an important role in determining complex VOCs such as polychlorinated biphenyls (PCBs) [113], trichloroethylene (TCE) [114], and benzene series [115] from contaminated soils. Zhang et al. [113] used SPME-PGC-MS to extract PCBs from soil matrices, achieving LODs of 10 ppm and RSDs around 20%. A coiled microextraction device coupled with portable GC-MS has been employed to detect 59 organic contaminants in soil samples [116]. Although these microextraction approaches

demonstrated lower quantitative accuracy and repeatability compared to benchtop instruments, they still provide valuable guidance for the qualitative and semi-quantitative analysis of soil VOCs. Wang et al. [114] designed a soil vapor sampling probe capable of efficiently collecting TCE from subsurface soil at various locations and depths. The analytes collected in the vapor collection bin were subsequently quantified in situ using SPME-PGC-MS. Vapor intrusion was further assessed by plotting three-dimensional distribution maps and concentration contours of TCE at different depths (Fig. 2d). Fatigante et al. [117] reported a modified PS coupled with portable mini-MS for detecting trace pesticide residues such as alachlor, aldicarb, aminocarb, buprofezin, and carbofuran in environmental soil samples, achieving detection of target agrochemicals as low as $\mu\text{g/g}$ levels. These methods have shown the utility of portable MS approaches in analyzing environmental contaminants in soil, they still require validation and standardization in practical applications, as well as comparative studies with traditional MS techniques, to ensure their accuracy and reliability, particularly given

the qualitative performances of many portable MS applications, since frequent absence of LOQ data in many studies.

3.4. Emergency and safety applications

In extreme environments, compared to laboratory MS, portable MS could offer unique advantages such as compact design, fast analysis, and minimal sample preparation, and thus has become a powerful analytical tool for rapid and reliable detection in emergency response, public safety, and forensic analysis. In environmental emergencies, portable MS facilitates real-time identification of contaminants from industrial accidents, chemical spills, and other hazardous releases, supporting risk assessment, evacuation planning, and targeted remediation. It has been widely reported that rapid detection of explosives and chemical warfare agents (CWAs) using portable MS. For example, Seto et al. [118–122] demonstrated onsite CWA identification using portable GC-MS. Bednar et al. [34] reported successful detection of munitions constituents in groundwater with portable GC-MS, where improved extraction and analytical methods enhanced identification accuracy and reduced false positives compared to conventional HPLC-UV. Recent studies also highlight onsite detection of non-volatile explosives such as trinitrotoluene (TNT) and perchloric explosives in smoke using portable mini-MS with EESI/SESI [71]. Beyond emergency response, portable MS has proven valuable in forensic investigations. Torres et al. [123] applied capillary microextraction coupled with portable GC-MS and needle trap devices to detect ignitable liquid residues and VOCs in fire debris, achieving detection limits around 10 ng for key compounds. Diaz et al. [124] used ULISSES, a compact 24 V-powered portable MS for in situ volcanic plume gas analysis under harsh conditions. Deployed at Costa Rica's Turrialba Volcano, ULISSES measured key gases including helium and sulfur dioxide, with airborne SO₂ data correlating closely with satellite observations. These applications highlight the portability of portable MS in extreme environments, yet also underscore the limitations of manual operation in highly inaccessible or hazardous areas. To address these challenges and extend analytical reach, the integration of portable MS with unmanned platforms has emerged as an attractive direction, enabling effective deployment in remote, hazardous, and hard-to-reach environments. For instance, Liu et al. developed MS robot systems for remote unmanned analysis of radioactive isotopes in contaminated zones (Fig. 2e) [71]. Overall, portable MS demonstrates broad versatility across routine environmental monitoring and high-risk emergency scenarios, enabling rapid onsite decision-making. Despite these advances, challenges remain in enhancing sensitivity, minimizing false positives/negatives in complex environmental matrices, and standardizing analytical protocols to ensure reproducibility and comparability. Continued efforts and innovation are highly expected to further promote portable MS as a critical tool for environmental emergency and safety monitoring.

4. Conclusions and prospects

Owing to the unique features of flexibility, portability, and convenience, portable MS has become an increasingly important tool in environmental analysis, enabling rapid, sensitive, and efficient detection and quantification of contaminants across a wide range of environmental matrices. The growing necessity for on-site monitoring and real-time analysis across various fields has significantly driven the rapid development of portable MS instruments at both academic and industrial aspects. This increased commercial availability reflects a rising demand for field-deployable instruments by end-users. However, portable MS generally does not yet match the performance of benchtop MS systems used in laboratories or monitoring stations [125]. This gap arises largely from the complexity of miniaturizing MS components, including ion source and interface, battery, mass analyzer, and vacuum system [23,24,26]. Achieving high-resolution and high-sensitivity of portable MS remains a huge challenge, especially for onsite analysis of

inorganic pollutants, heavy metals, and microorganisms. Despite these challenges, an ideal portable MS system for environmental applications should incorporate several key features: 1) compactness, low energy consumption, rapid analysis speed, and user-friendly operation to enable immediate and efficient field deployment; 2) advanced data processing capabilities, including artificial intelligence integration, to support fast and accurate analyte identification, particularly for toxic and harmful unknown pollutants; 3) stability and robustness in complex, harsh, or remote environments, as well as the capability for deployment via unmanned robotic platforms. Overall, new technological development is enhancing the effectiveness and accessibility of portable MS, positioning it as a vital tool to meet requirements of current and future environmental analysis.

CRedit authorship contribution statement

Jianfeng Zhang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Zhen Zhou:** Writing – review & editing, Supervision, Resources, Project administration. **Bin Hu:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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